

Chapter 16: Natural Resources

Textbook Exercise Solutions & Answers



1. Describe natural resources with reference to the following 3 types:

(a) Mineral resources

Mineral wealth has an important place among natural resources. The rocks on the earth are mainly made of minerals. These minerals can be obtained by mining. Minerals that contain a high proportion of metals are ores, (i) Minerals are formed from the magma in the earth's crust and the lava from the eruption of volcanoes when they cool and get transformed into crystals e.g. magnetite, mica, (ii) Minerals get transformed from one form into another due to large changes in temperature and pressure e.g. Diamond, Graphite, (iii) There are 3 types of minerals according to their properties Non-metal minerals: e.g. mica, sulphur, potash, diamond.

- Metals minerals: e.g. Iron, gold, silver, tin, bauxite, platinum.
- Energy minerals: e.g. Coal, mineral oil, natural gas, diamond, ruby, sapphire, jade are used as gems.

Some important minerals and ores are Iron ore, manganese, bauxite, copper and mica.

(b) Forest resources

1. A forest is a natural habitat of plants, animals and microbes.
2. Forest perform certain specific protective and productive functions.
3. We get different types of wood and many medicinal plants from forests.
4. Forest wealth includes fire wood, trees like teak, mahogany, neem, acacia, subabhl, fibers, paper, rubber, gum, aromatic substances.
5. We get fragrant essential oils from lemon grass, vanilla, kewada, vetiver (khus) and eucalyptus.
6. Sandalwood and oil of eucalyptus are used for making soaps, cosmetics and incense sticks.
7. Also we get fruits, bulbs, roots, honey, sealing wax, dyes from forest.
8. Many medicinal plants adalsa, bel, neem, periwinkle, cinnamon, are used for treatment of various diseases such as cough cold, malaria, diarrhoea, cancer, fever and cold.
9. Wood is used as a source of energy for cooking purpose and for keeping warm.
10. For making furniture, tool handles, matches, bridges, boats etc.
11. Bamboos are used for matting, flooring, basket, ropes, rafts, cots etc.
12. Bamboos, wood are used in the manufacture of rayon, yams, artificial silk-fibers.

(c) Ocean resources

1. Oceans occupy about 70% of a earth's surface.
2. Energy can be obtained on a large scale from oceans.
3. Sea waves at high and low tide and ocean currents are being used for generation of power.
4. There are billions of tons of minerals dissolved in ocean water.
5. There are large reserves of tin, chromium, potassium, phosphates, copper, zinc, iron, lead, thorium, manganese, sulphur, uranium in the ocean and sea beds.
6. Also we get many types of gems, conches, shells and pearls from the sea.
7. There are large reserves of mineral oil and natural gas at the bottom of sea.
8. We get fishes, dried shrimp, shells, fungi, shark, cod fish and sea cucumber from the ocean.

2. Write answers to the following questions in your own words.

(a) What is meant by fossil fuel? What are their types?

1. Fossil fuels (such as coal or natural gas) are formed in the earth from dead plants and animals, due to high pressure from above and the heat from the earth's interior.
2. The dead plants, animals get transformed into fossil fuel.
3. There are three types of fossil fuels which can be used for energy production, Coal, oil and natural gas.

a. Coal:

1. Coal is a solid fossil fuel formed over millions of years by the decay of land vegetation,
2. Coal is a store house of carbon. Coal is found in mines,
3. It is like porous, hard, black rock.
4. It is burnt to obtain heat energy.
5. It is used as a fuel in thermal power plants and to run boilers and railway engines, for cooking and baking bricks.
6. Anthracite is the coal of the highest grade.
7. Producer gas and water gas are obtained from coal.

b. Mineral oil:

1. Mineral oil is the liquid fuel formed by the decomposition of organic substances buried underground,
2. The underground mineral oil is extracted through oil wells.
3. Mineral oil is found mainly in oil sands, shale, sandstone and limestone at a depth of about 1000 to 3000 metres.
4. Mineral oil is also known as petroleum or crude oil. It is greenish-brown in colour.
5. Petroleum is a mixture of many compounds mainly of the hydrocarbon type.
6. It also contains compounds of oxygen, nitrogen and sulphur.
7. Petroleum is extracted through oil wells and refined by fractional distillation to separate other components,
8. Aviation petrol, gasoline, diesel, kerosene, naphtha, lubricating oil, tar are all obtained from petroleum.
9. They are used as fuel and for production of dyes, pesticides, perfumes and artificial fibres.

c. Natural gas:

1. Natural gas is an important fossil fuel.
2. It is lighter than air. It is found with petroleum in underground oil wells and in some places as natural gas alone.
3. The main component of natural gas is methane (CH_4), ethane (C_2H_6), propane (C_3H_8) and butane (C_4H_{10}) are present in small proportions.

4. This gas is highly flammable.
5. This fuel can be carried over long distance by means of a gas pipeline.
6. In absence of pipeline, it is transformed under high pressure into compressed natural gas (CNG) and liquefied natural gas (LNG).

(b) Make a list of the components we obtain from mineral oil.

1. Mineral oil is the liquid fuel. It is also known as petroleum or crude oil. Petroleum is refined by fractional distillation to separate other components.
2. Aviation petrol, gasoline, diesel, kerosene, naphtha, lubricating oil, tar are the components we obtain from mineral oil.

(c) What do we get from forest?

1. A forest is a natural habitat of plants, animals and microbes. We get wood, firewood from trees like teak, mahogany, neem, subabul.
2. Also we get fibers, paper, rubber, gum and aromatic substances.
3. We get many medicinal plants like Adulsa, Bel, Neem, Periwinkle, Cinnamon, Cinchona which are used for treatment of various diseases like cough, cold, diarrhea, fever, cancer, diarrhoea, nausea, malaria.
4. We get fragrant essential oils from lemon grass, vanilla, kewada, khus, eucalyptus.
5. In addition to this we get fruits, bulbs and roots, honey, sealing wax, catechu, dyes from forest.

(d) What are the items included in ocean resources? What are their uses?

1. Tons of minerals are found in ocean water.
2. They are tin, chromium, phosphates, copper, zinc, iron, lead, manganese, sulphur, uranium etc.
3. We also get many types of gems, conches, shells and pearls from the sea.
4. Mineral oil and natural gas is found at the bottom of sea.

Bio-resources in oceans

- Fishes – like pomfret, seer fish as also shrimps
- prawns – they are sources of proteins and vitamins, therefore, mainly used as sea food
- Dried shrimp, Bombay duck powder – used as poultry feed and is a good manure.
- Shells – used for preparation of medicines, ornaments and decorative articles.
- Fungi – used for the production of antibiotics.
- shark and cod fish – used for producing edible oil rich in Vitamins A, D and E.
- Sea cucumbers – used as medicine for treating cancer and tumours.

Mineral resources from oceans

- Thorium – used in the production of atomic energy.
- Magnesium – used in the flash bulb of a camera.
- Potassium – the main ingredient in production of soap, glass, fertilizer.
- Sodium – used in the production of cloth and paper.
- Sulphate – used in making artificial silk.

(e) Why should we prevent the wastage of fuels used for vehicles?

1. Vehicles are powered by gasoline or diesel petrol.
2. Nowadays compressed natural gas (CNG) is also used.
3. Aviation petrol, gasoline, diesel, we get from petroleum after fractional distillation.
4. Petroleum is a non-renewable source of energy.
5. It is getting depleted day by day and the demand for fuel has increased greatly due to the rapidly growing population.
6. But the reserves of petroleum (fossil fuel) are limited.
7. It is becoming difficult to meet the increased demand. So we should prevent the wastage of fuel.

(f) Why is the diversity of plants and animals in the forests declining?

1. The major causes of biodiversity decline are land use changes, pollution, changes in atmosphere, CO₂ concentration, changes in the nitrogen cycle, acid rain, climate alteration and introduction of exotic species.
2. All this is due to population growth.
3. This is the main cause of the loss of biodiversity. More of wood for fuel is been used, more land is required for food production, livestock grazing etc.
4. Rainforest land is converted to industrial and residential usage, so natural habitat of birds, animals is destroyed.
5. Due to population growth most of wild land is converted to crop land. So no space for animals to live, their natural habitat is destroyed.

(g) Write the names of five minerals and the useful substances obtained from them.

Some important minerals and ores are:

1. Iron ore: It is used for making farming implements, rails of railway tracks etc.
2. Manganese: Compounds of manganese are used in the preparation of medicines and for giving a pink tinge to glass. Manganese is also used in electrical appliances.
3. Bauxite: It contains 55% Aluminium. Therefore, it is used mainly in aeroplanes, transport vehicles and to make electric wire.
4. Copper: It is used to make electric wires as well as in radios, telephones, vehicles, and for making kitchen utensils and statues.
5. Mica: Mica has many uses such as in ayurvedic medicines, dyes, electric machines and equipment, wireless communication equipment, etc.

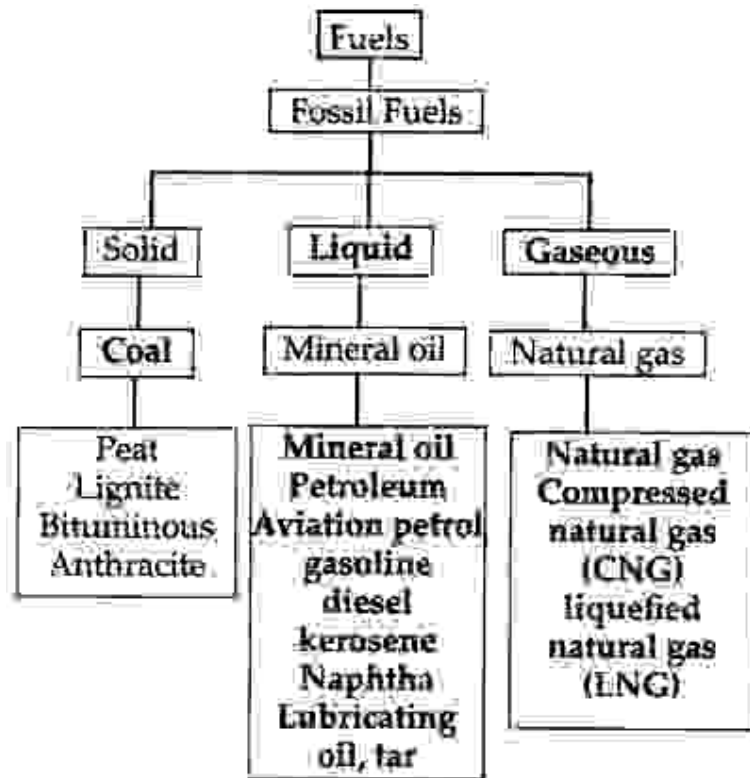
(h) Name the two important stages in the process of obtaining metals from ores.

The two important stages in the process of obtaining metals from ores are extraction and purification.

3. What steps are taken for protection and conservation of natural resources?

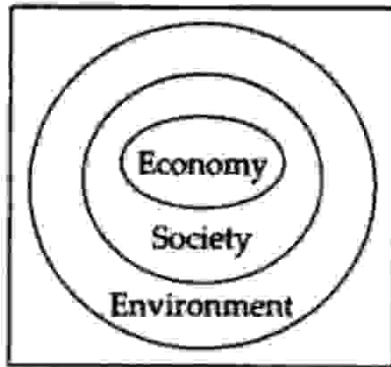
1. Protection and conservation of natural resources is necessary to protect the natural world. (CNRM) Community Natural Resource Management has been promoted in recent years for biological conservation.
2. The World Conservation Union has been formed. It is the world's leading authority and democratic platform in conservation and sustainability.
3. Young trees should not be cut. More and more trees need to be grown.
4. The stringent restrictions/ laws/ Regulations regarding use of forests should be strictly followed. Awareness is being created through Mass media.

4. Complete the Flow Chart:



5. How does the economic condition of a nation depend on its natural resources?

1. Natural resources are a necessary condition for economic growth.
2. Three circles enclosed within one another shows how both economy and society are subsets of our planetary ecological system.



3. Natural resource economics deals with the supply, demand and allocation of the Earth's natural resources.
4. Economic growth is an increase in the value of goods and services produced in an economy.
5. The natural resources of a country depend on the climatic and environmental conditions.
6. Countries having plenty of natural resources enjoy good growth than countries with small amount of natural resources.
7. A country having skilled and educated work force with rich natural resources takes the economy on the growth path, because skilled and educated people can efficiently utilize or exploit natural resources, e.g. Saudi Arabia: their economic growth is high, because they have oil wells.
8. All countries import fuel or oil from them so Saudi Arabia is economically a rich country.

6. Which medicinal plants will you grow on your school premises and near your house? Why?

1. I will grow tulsi, neem, lemon grass, bel, adulsa, periwinkle, cinnamon, ashwagandha, shatavari, amla, hirda, behda.

2. These plants are used for treatment of various diseases.

- Tulsi → for cough and cold.
- Amla, hirda, behda → Help in digestion of food or any stomach related problems.
- Adulsa → for cough and cold
- Bel → for Diarrhoea
- Neem → Fever and cold
- Periwinkle → Cancer
- Cinnamon → Diarrhoea, nausea.
- Cinchona → Malaria.
- Ashwagandha → It can reduce blood sugar levels. It can reduce stress and anxiety, may reduce symptom of depression.
- Shatavari → has been used for centuries in Ayurveda to support the reproductive system.
- Amla → Excellent source of vitamin C. Amla juice is best tonic to keep you younger for long.
- Hirda → This fruit is commonly called Haritaki. It is used for treating arthritis, dental problems like caries, bleeding gums.
- Behda → (Beleric) is a rejuvenative and laxative, proves beneficial for hair, throat and eyes.
- Tulsi → Tulsi has germicidal, fungicidal, anti-bacterial anti-biotic properties, cures fever. Treats diabetes, protects the heart. Tulsi has anti-oxidant component Eugenol which keeps one's blood pressure under control.